Exploring Intersectionality in Social Media: An Approach to Understanding the Double Marginalization of Minority Women in the Global South

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Abstract

The marginalization of people belonging to minority communities is a worldwide issue. Women, in general, living in the Global South already face oppression for existing societal reasons. Women here who also belong to any minority community face double marginalization, or in better words, intersectionality for (a) belonging to the community and (2) for simply being women. In this work, I study the unique challenges of intersectionality faced by minority community women in the Global South while using social media. Through a qualitative approach and synthesizing quantitative patterns with prior HCI and global south literature, here, I explain how minority women living here face intersectionality.
Keywords: Intersectionality, Double Marginalization, Social Media, Hate Speech, Ethical AI Human Computer Interaction (HCI), Human-Centered Computing.

1 Introduction

Marginalization of minority communities is still a global issue to date, which is shaped by social, cultural, and structural inequalities. Women often face gender-based discrimination, which is also not a new issue. When these two get combined, they create intersectionality for women belonging to minority groups. Social media has a significant impact on everyone's lives these days. Even though these impacts are positive for the majority of people, they can often bring negative consequences on people who are at the intersection of overlapped social barriers. In this study, I analyze two publicly available datasets to prove that minority women face

intersectionality while using social media. And I also perform meta-analysis on existing literature to address this existing underestimated social issue. By analyzing the patterns of hate speech and discourse in the context of different groups, this research applies pattern recognition techniques to reveal how the experiences of women belonging to a minority community on social media differ from those of women in general or from men belonging to the same community.

2 Related work

2.1 Intersectionality through the lens of Pattern Recognition

Traditional pattern recognition and machine learning approaches often rely on only one aspect rather than capturing every possible aspect, which can often miss out on intersectionality. Prior works have shown how models work fairly when evaluated separately across attributes like gender or race, but still may produce different outcomes for intersectional subgroups, including minority women [1]. Kong [2] discusses the rigorous approach to intersectional fairness in AI algorithms, which focuses on getting statistical parity across intersectional subgroups defined by combinations of protected attributes like race, gender, and social class. The paper talks about three main issues with this approach- it is overly focused on identity categories, like race and gender, without paying attention to underlying problems like sexism or racism. Secondly, it establishes a conundrum where either we continue to break down groups into smaller and smaller subgroups to the point of infinite regress, or we arbitrarily select which groups to prioritize among, which can be seen as being unfair. And thirdly, the paper argues that it is far too narrow to think of equality simply through the lens of statistics. The paper argues that what is needed is a much more robust and nuanced method of thinking about fairness, and that we should move towards a kind of nondistributive understanding of fairness.

2.2 HCI in intersectionality

HCI has a long history of working with marginalized communities. A study discussed by Schlesinger et al. [3], explores the identity of users using intersectionality as a framework. It examined 140 papers published in the CHI conference between 1986 and 2016. The study

mainly focused on gender, class, and race. The paper argues that intersectionality is often overlooked by researchers, as HCI researchers mostly examine one identity at a time. This paper mainly emphasizes why researchers should lean more towards examining intersectionality in this field. Another study by Erete, Rankin, and Thomas [4] addresses intersectional analysis of power as a way to call out systemic oppression in HCI and design. The authors argue that traditional HCI research fails to handle the historical context of structural oppression. The study uses intersectionality as one of the frameworks.

2.3 Intersectionality in the Global South

Intersectionality in the global south creates a unique form of discrimination. A paper by Jahan [5] addresses the condition of Muslim women in India. Women there face discrimination for belonging to a minority community and more discrimination for being a woman. This keeps them in a more complex state than Muslim men living there. A study by Munro, Parker, and McIntyre [6] discusses the intersectionality faced by Bangladeshi Garo women, especially their food insecurity. The intersectionality framework is applied to understand the dynamics of overlapping factors like gender, poverty, and indigeneity. The article brings to attention the challenges these women experience with regard to access to culturally suitable food as well as the socio-political circumstances they are exposed to.

3 Methodology

I conducted the study in three steps. At first, I went through the existing literature in this field. I went through various types of research to do the meta-analysis. Then, for the second step, I took two publicly available datasets related to hate speech on social media and analyzed them. And finally, I interviewed two people belonging to a minority group in Bangladesh to know about their experiences regarding social media usage.

3.1 Meta-Analysis

For this part, I went through three types of studies. Firstly, I analyzed studies that work with intersectionality issues with a pattern recognition approach. Secondly, I studied how intersectionality is perceived with the HCI approach. And finally, I inspected the existing social studies about the issue in the Global South. The first two types revealed how underexplored this field is in human-centered computing, and the third type confirmed the existence of double marginalization.

3.2 Analyzing Datasets

I used two publicly accessible datasets from Kaggle and Hugging Face to analyze them and show the proof of concept. The first dataset is used to show the actual scenario of intersectionality globally. As getting any dataset based on the Global South was not possible for the project, I used the available one just to give an idea of the issue. The second dataset is a dataset of annotated hate speech. This is used to show how different people perceive hate speech differently.

Colab Notebook: CSE424_Project_Final.ipynb

The first one is the HateXplain dataset. I used the processed and labeled version of the dataset named final_hateXplain.csv. This contains comments labeled as normal, offensive, or hate speech. It also has attributes like race, religion, gender, and sexual orientation. The target of this analysis was to explore whether minority women experience a higher rate of hate compared to general women. For data pre-processing, I cleaned the column names to remove any leading or trailing spaces. Then I created a new binary variable named ‘is_minority’ to identify if a person belongs to a minority group based on ‘Race’ or ‘Religion’. This variable is defined as True if either Race or Religion is specified as something other than "No_race" and "Nonreligious", respectively. For segmentizing the data, two subgroups were created based on gender: Women and Minority Women. To categorize the hate speech, I created two subsets of the Women and the

Minority Women groups as Women Hate Speech and Minority Women Hate Speech. Using the formula below, I calculated the hate speech rate,

Hate Speech Rate = Number of Hate Speech Records / Total Number of Records in Group

After this, to compare the rates, I created a bar chart showing the proportion of hate speech targeting Women and Minority Women. Also, a stacked bar chart was generated to visualize the composition of Hate versus Non-Hate speech within each group. The Chi-square test results indicated whether the differences in hate speech rates between the two groups were statistically significant, providing insights into whether Minority Women face higher levels of hate speech than Women overall.

The second dataset that I used is the UCberkeley measuring-hate-speech dataset. I downloaded the available CSV file and analyzed it. The dataset has multiple attributes related to hate speech annotations. This datasheet is basically a selection of hate speech that is annotated by various people. It is decided by people if the comments are offensive or not, and to whom they are targeted. For preprocessing and cleaning, I did the same column stripping. The data were aggregated at the comment-level (one record for every comment) using either the mean of hate speech scores (e.g. hate_speech_score/hatespeech) and then the first entry for the text variable (as applicable). Further, binary variables for target_gender_women, target_race, target_religion, etc., were created for each record using the maximum value for each record from multiple annotators. For defining group variables like is_woman_targeted and is_minority_rr were defined. Then, a threshold of 0.5 was used to classify comments as hate speech or non-hate speech based on the hate_speech_score. Comments with a score greater than 0.5 were classified as hate speech. For statistical analysis, a chi-square test of independence was performed to show the difference in proportions of hate speech between women and minority women. Then I generated a bar chart and a histogram to visualize everything.

3.3 User Interviews

For various limitations, it was not possible to conduct interviews and surveys on a broader scale. But I was able to interview two people from one of the minority communities in Bangladesh. One of our interviewees was a man, and another was a woman. Both shared their experiences with social media while belonging to a minority community in Bangladesh.

4 Interpretation

The meta-analysis proves that traditional machine learning fails to capture intersectionality but works really well with one-dimensional data. Even though HCI is the only computer science field that works with societal issues directly, it does not have enough work on double marginalization. Also, all the papers in this field regarding intersectionality are quite old. The social analysis studies of the Global South prove the existence of minority women being subjected to double marginalization.

The HateXplain dataset analysis proves that minority women are targeted with more hate than women in general on social media. The hate speech rate for minority women is higher than for women overall, indicating that minority women are more likely to be the subject of hate speech online.

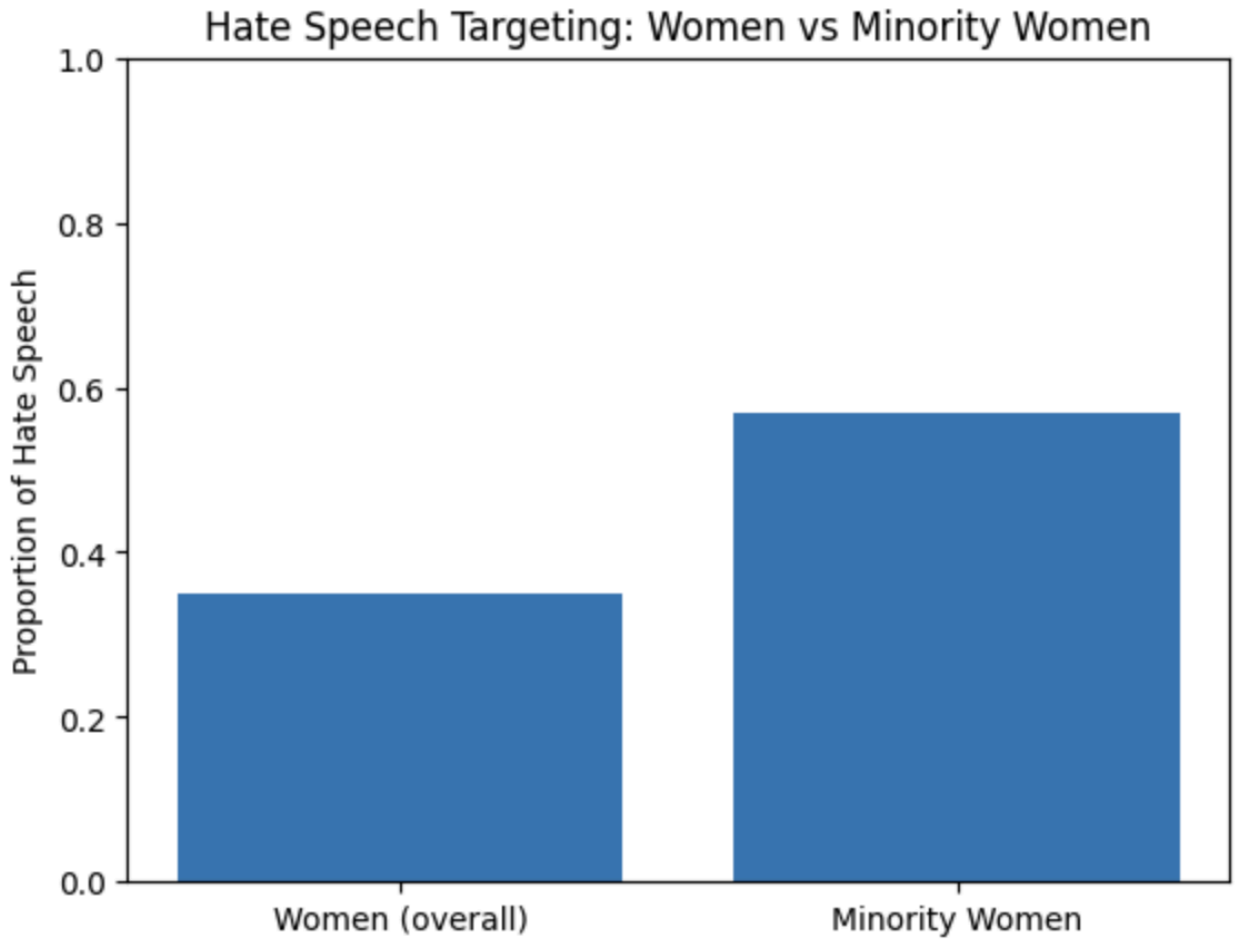

Figure 1: Proportion of hate speech targeting Women vs Minority Women (HateXplain dataset)

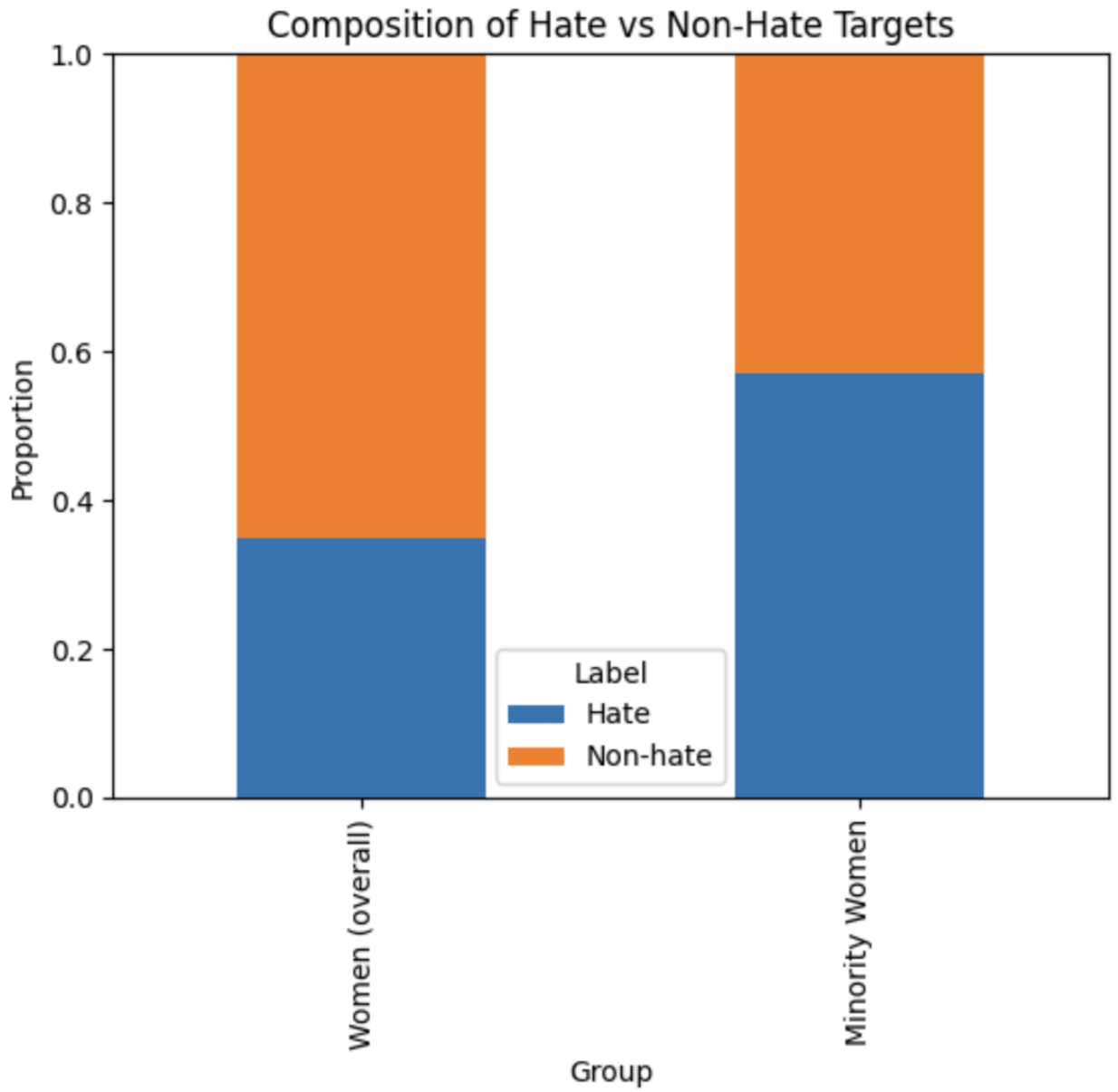

Figure 2: Composition of hate vs non-hate for Women overall and Minority Women (HateXplain dataset)

The UCberkeley dataset shows how minority women's and women overall's experiences with hate speech might differ in terms of how they perceive or experience hate speech. This aspect is particularly important in understanding the visibility and nature of the hate targeted at them. Even though all of my analysis confirms that minority women face double marginalization, for

the nature of this datasheet, it shows that minority women experience slightly less hate speech compared to women overall. This is the difference in perception. The frame of impact of this dataset also says the same thing.

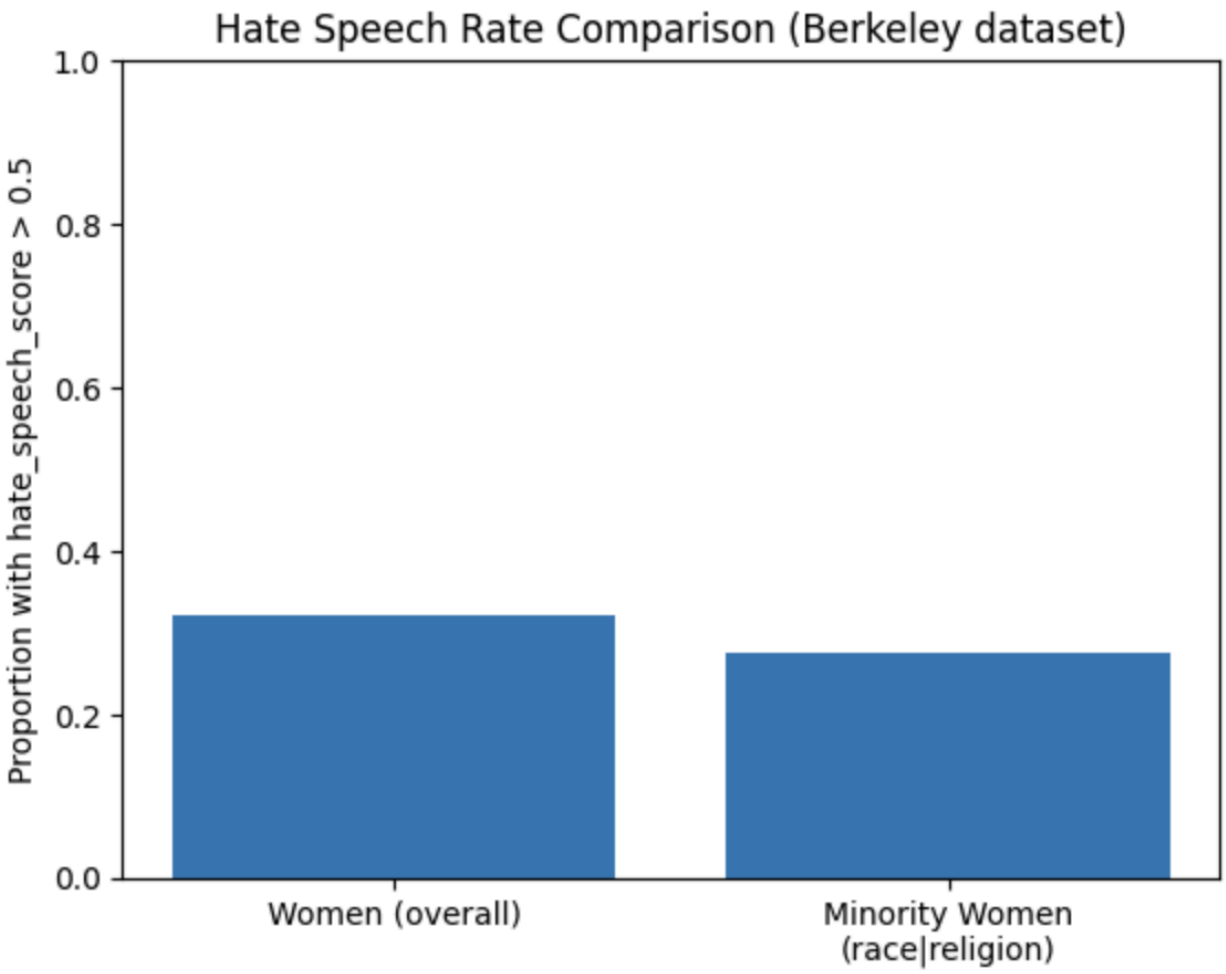

Figure 3: This figure shows the difference of perception (UCberkeley dataset)

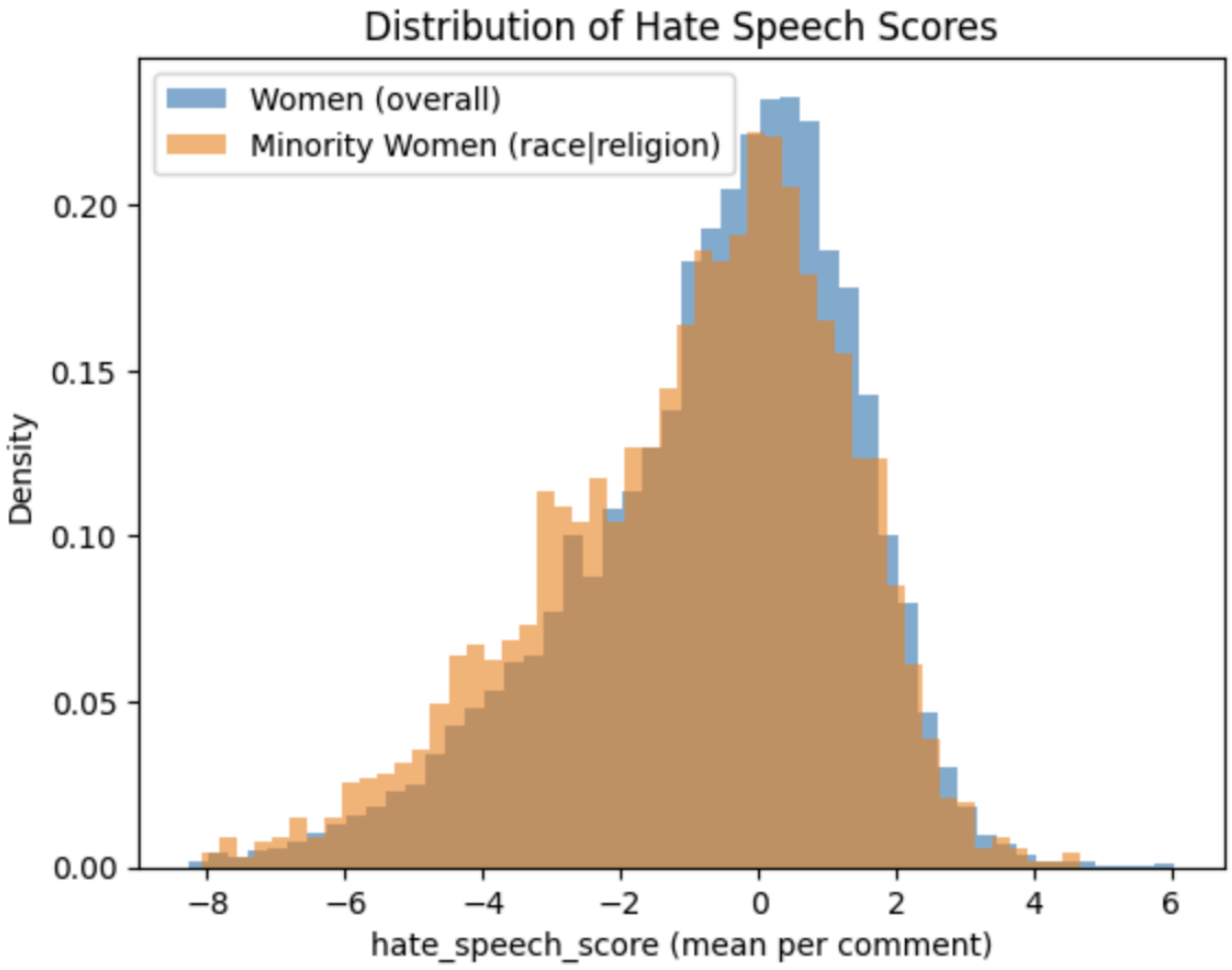

Figure 4: Distribution of Hate Speech Scores for Women (overall) and Minority Women (race|religion) (UC Berkeley dataset)

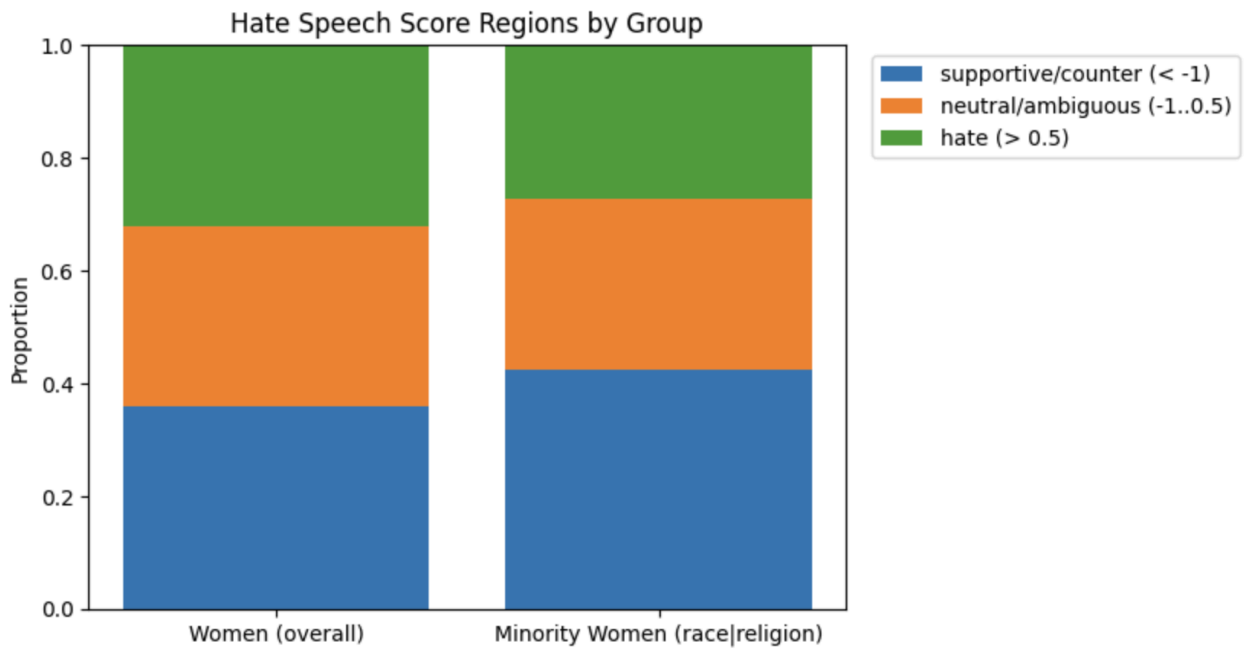

Figure 5: Comparison of Hate Speech Rates between Women (overall) and Minority Women (race|religion) based on the hate_speech_score threshold of 0.5 (UC Berkeley dataset).

The bar chart, the distribution, and the comparison all point to the difference of perception. As the datasheet is annotated by all kinds of people, the hate speech towards minority women is perceived differently. This is why, even though it is very obvious that minority women are subjected to more hate speech, it is not properly represented in the datasheet.

From the interviews, both interviewees talk about how, for the same reason, when one uses social media is useful for general people, it has adverse effects on them. The male interviewee at one point states, “As men, we have to be cautious; if we have to be cautious, then of course our women have to stay way more cautious than us while using social media". And the female interviewee confirmed this case too. She stated how they use social media anonymously sometimes to talk about their faith.

All my findings point to the fact that women belonging to minority communities face marginalization not only for being in that group, but also for simply being a woman. Human-centered computing needs to have more work in this field. At the same time, it is very

noticeable from the data analysis that this is a worldwide issue, but due to differences in perception, normal people might often overlook it.

5 Limitations and future work

The limitation of my research is mainly the lack of a Global South-based dataset. This work is just a beginning. For future work, we can refine this study with a proper dataset and conduct interviews on a broader scale. Then we can build up our work with traditional pattern recognition methods and HCI approaches. This is just the beginning of a very large scope of human-centered computational work.

6 Conclusion

This research investigates intersectionality and its impacts on minority women's experiences on social media through studying double disadvantage through both gender and minority status. Using both human-computer interaction and social science literature reviews as well as analysis of the HateXplain Dataset and UC Berkeley's Measuring Hate Speech Dataset, the conclusion of this analysis reveals that minority women indeed endure more intersections of disadvantage than women alone on social media. Furthermore, through analysis of both HateXplain and UC Berkeley's analysis of user-delineated hate speech, it is illustrated that minority women endure more instances of hate speech and that the values of human interpretation and opinion often interfere with precise representation. Lastly, it is supported that minority women modify their behavior on social media through greater personal awareness and personal anonymity.

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AI Chat 1: https://gemini.google.com/share/44fa89ceb430

AI Chat 2: https://chatgpt.com/share/695eb3f9-7a40-800d-aaca-2f9ec697da6a